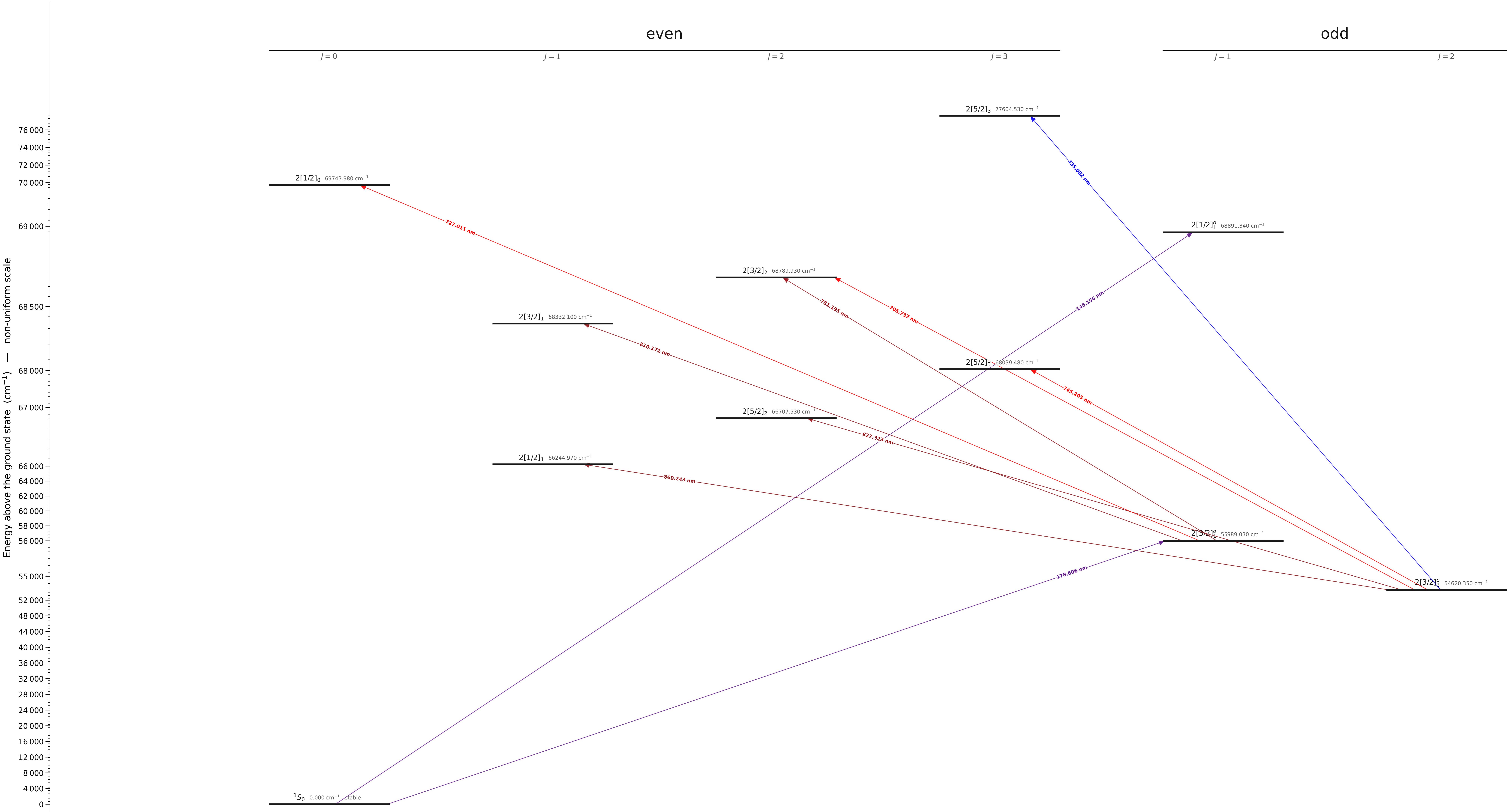
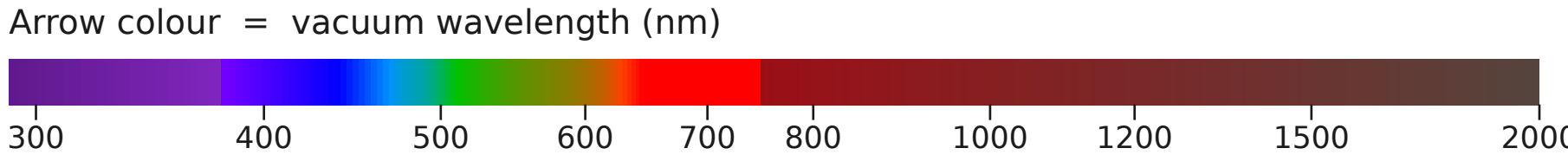


Radon

Rn (Rn I). Strongest classified transitions below 2 μm (NIST): 11 levels, 10 lines.

Rn⁺ ionisation limit 86692.50 cm⁻¹ = 10.74850 eV (115.350 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle J \| e r \| J' \rangle |$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$ \langle J \ e r \ J' \rangle $ (ea_0)	A (s ⁻¹)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$2[3/2]_2^o - 2[5/2]_3$	745.2048	744.9996	402.2954	-	-	-	-	the line of the only laser-spectroscopy study of radon: c...

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.

²²¹Rn hyperfine structure ($I = 7/2$)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$2[3/2]_2^o$	-18.4(6)	2184(4)	F=3/2: +1335.81 F=5/2: +314.64 F=7/2: -568.91 F=9/2: -768.74 F=11/2: +417.30
$2[5/2]_3$	-3.8(4)	2082(6)	F=1/2: +1166.25 F=3/2: +788.75 F=5/2: +258.75 F=7/2: -275.00 F=9/2: -604.25 F=11/2: -461.25 F=13/2: +481.25

Measured A, B constants; sources in the data file.